Project Summary: Trader Behavior & Market Sentiment Analysis

1. Methodology

This analysis evaluates the impact of Bitcoin market sentiment—derived from the Fear & Greed Index—on trading performance on the Hyperliquid exchange. The pipeline was structured into three phases:

- **Data Integration:** Merged high-frequency trade logs with daily sentiment scores, utilizing .dt.normalize() to align disparate temporal datasets.
- **Behavioral Clustering:** Employed **K-Means Clustering** to segment traders based on activity frequency, average position size, and win-rate consistency, creating actionable "trader archetypes."
- **Predictive Analytics:** Implemented a **Random Forest Classifier** to assess the probability of next-day profitability using sentiment-encoded features and historical performance metrics.

2. Key Insights

- **Sentiment Sensitivity:** Data suggests that aggregate win rates are inversely correlated with "Greed" sentiment phases, indicating that traders often fall victim to over-leverage or "FOMO" during bull cycles.
- **The "Frequent Trader" Trap:** Behavioral analysis reveals that traders in the "Frequent" archetype exhibit lower net PnL per trade compared to "Infrequent" cohorts. This suggests that high-turnover strategies on this platform frequently erode capital via slippage and over-trading.
- **Resilience Archetypes:** A specific segment of "Consistent Winners" showed higher resilience during "Fear" cycles, indicating a reliance on mean-reversion and disciplined entry points during market volatility.

3. Strategic Recommendations

Based on the evidence, the following rules of thumb are proposed for algorithmic risk management:

1. **Sentiment-Adaptive Risk Scaling:** Automatically reduce individual trade position sizes by **25%** when the daily sentiment classification shifts to "Greed." This acts as a circuit breaker for accounts prone to emotional trading during market peaks.
2. **Activity Capping for High-Turnover Accounts:** For traders categorized as "Frequent" (based on daily volume metrics), implement a daily execution cap of **[X] trades per day**. This forces a transition from reactive, high-turnover execution toward more deliberate, high-conviction setups, which statistically correlate with higher win rates in this dataset.
3. **Mean-Reversion During "Fear" Cycles:** Shift automated execution parameters to prioritize support-level accumulation during "Fear" cycles, as historical data shows superior risk-adjusted returns when entering positions while market sentiment is depressed.